

COSC 594/650: Optimization for Machine Learning

Homework 1

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This homework is due on **9/4, 11:59pm**. Format your solution in PDF and submit it on Canvas. You only need to submit your answers to 5 out of 6 problems in order to receive full credit. If you answer all 6 problems, the extra one will be considered as a bonus.

1. (**Convex optimization**) Consider the optimization problem

$$\begin{aligned} & \underset{x}{\text{minimize}} && e^{x_1} + e^{x_2} + \cdots + e^{x_n} \\ & \text{subject to} && x_1 + x_2 + \cdots + x_n = \gamma. \end{aligned}$$

- (a) Show that the above optimization problem is convex.

Solution:

- (b) Use the equation

$$x_n = \gamma - x_1 - x_2 - \cdots - x_{n-1}$$

to rewrite the above optimization problem as an unconstrained optimization problem.

Solution:

- (c) Find the set of global optimum points.

Solution:

- (d) Apply the change of variables $y_i = e^{x_i}$ to prove the arithmetic-geometric mean inequality

$$\frac{1}{n} \sum_{i=1}^n y_i \geq \left(\prod_{i=1}^n y_i \right)^{1/n}, \quad \forall (y_1, \dots, y_n) \in \mathbb{R}_+^n.$$

Solution:

2. (**Convexity proofs**) Prove the following statements.

(a) The function

$$f(x) = \log \left(e^{a_1^T x + b_1} + \dots + e^{a_n^T x + b_n} \right)$$

is convex.

Solution:

(b) Let $\sigma_1(Z)$ denote the largest singular value of the matrix Z . Show that the function

$$f(X) = \exp \left((\sigma_1(AX))^{10} \right)$$

is convex.

Solution:

3. (**Stationary points**) For each value of the scalar β , find the set of all stationary points of the optimization problem

$$\begin{aligned} & \underset{x,y}{\text{minimize}} && x^2 + y^2 + \beta xy + x + 2y \\ & \text{subject to} && (x,y) \in \mathbb{R}^2. \end{aligned}$$

Which of these stationary points are global minima? For an unconstrained smooth optimization problem, a point is stationary if the gradient of the objective at that point is zero.

Solution:

4. (**Linewise local minima**) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable. Suppose that x^* is a local minimum of f along every line that passes through x^* ; that is, the function

$$g(\alpha) = f(x^* + \alpha d)$$

has a local minimum at $\alpha = 0$ for every $d \in \mathbb{R}^n$.

(a) Show that $\nabla f(x^*) = 0$.

Solution:

(b) Show by example that x^* need not be a local minimum of f . *Hint:* Consider

$$f(y, z) = (z - py^2)(z - qy^2), \quad 0 < p < q.$$

Show that $(0, 0)$ is a local minimum of f along every line that passes through $(0, 0)$. Furthermore, if $p < m < q$, then $f(y, my^2) < 0$ whenever $y \neq 0$, while $f(0, 0) = 0$.

Solution:

5. **(Symmetric rank-one matrix approximation)** Let $A \neq 0$ be a symmetric rank-one square matrix; that is,

$$A = x_0 x_0^T \neq 0, \quad x_0 \in \mathbb{R}^n.$$

Consider the optimization problem

$$\begin{aligned} & \underset{x}{\text{minimize}} && \frac{1}{2} \|A - xx^T\|_F^2 \\ & \text{subject to} && x \in \mathbb{R}^n. \end{aligned}$$

- (a) Is this optimization problem convex?

Solution:

- (b) Find the set of stationary points; that is, find all $\bar{x}$ for which $\nabla f(\bar{x}) = 0$, where f is the objective function.

Solution:

- (c) Is every local optimum of the problem a global optimum?

Solution:

6. **(Rank-one matrix factorization)** Let $A \neq 0$ be a rank-one square matrix; that is,

$$A = x_0 y_0^T \neq 0, \quad x_0, y_0 \in \mathbb{R}^n.$$

Consider the optimization problem

$$\begin{aligned} & \underset{x,y}{\text{minimize}} && \frac{1}{2} \|A - xy^T\|_F^2 \\ & \text{subject to} && x, y \in \mathbb{R}^n. \end{aligned}$$

- (a) Is this optimization problem convex?

Solution:

- (b) Find the set of stationary points.

Solution:

- (c) Show that every local optimum of the problem is a global optimum.

Solution: